

5.1 Architectural Substrate



Explicit Context



- Dense attention
- Sliding windows
- Sparse attention

Compressed State



- Linear Attention
- Recurrent/SSM Memory
- Latent KV memory

Hybrid Architecture



- Attention-state hybrids
- Global-linear paths
- Block-wise composition

High-Throughput



- Speculative decoding
- Multi-token prediction
- KV-cache optimization



5.2 Data & Envs Synthesis

Task Synthesis



- Dependency chains
- Graph/ State Structure
- Adaptive Curricula

Environment Synthesis



- Symbolic environments
- Neural world models
- Neural-symbolic hybrids

Trajectory Synthesis



- Env-grounded records
- Policy rollouts
- Failure Reuse



5.3 Pre/Mid-training



Reasoning

- Code & math priors
- Agent data mid-training



Long Context

- long-context training
- Position scaling



Multimodal

- Vision-language grounding
- Omni-modal training



Data Mixture & Strategy

- Capability-Oriented Corpora
- Domain-Mixture Optimization



5.4 Fine-tuning

Instruction Selection



- Quality judging
- Instruction mixing

Curriculum Learning



- Horizon curriculum
- Difficulty curriculum
- Tool-complexity curriculum

Trajectory Distillation



- Strong-to-weak distillation
- Process-aware distillation
- Self-distillation



5.5 Agentic Reinforcement Learning



Credit Assignment

- Outcome credit
- Process credit
- Rubric signals



Policy Optimization

- Clipping-based Opt.
- Turn-level Opt.
- Entropy-based Opt.



Sampling Strategy

- Trajectory rollouts
- Tree sampling
- Budget-aware pruning



Interaction Patterns

- Hierarchical control
- Multi-agent coordination
- Memory-state learning



5.6 On-Policy Distillation

Teacher-guided OPD



- Turn-level interpolation
- Step-wise supervision
- Multi-teacher guidance

Self-improving OPD



- Privileged signals
- Environment feedback
- Failure Reflections

5.7 Self-Evolution

Offline Self-Evolution



- Generate-Filter-Train
- Offline Experience Curation
- Experience write-back

Online Self-Evolution



- Online Experience Collection
- Continual Policy Update
- Self-play curricula

Agent-Environment Co-Evolution



- Environment tuning
- Difficulty adaptation
- Environment-task discovery